SOFTWARE REQUIREMENTS SPECIFICATION

Software Requirements Specification (SRS)

Agri-Energy Desk Digital Platform: Version 2.0, revised to reflect the as-built platform, for NREP review and approval

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional requirements for the Agri-Energy Desk Digital Platform. Version 1.0 was written before development and described the platform to be built. This revision restates those requirements against the platform as it now stands: every requirement below has been implemented and is available in the User Acceptance Testing environment, except where Section 10 records otherwise. It remains the basis for design, development, testing and acceptance, and it is submitted to NREP for review and approval as the current definition of the system.

1.2 Scope

The Agri-Energy Desk is a national digital knowledge management and collaboration platform for Uganda's Agri-Energy sector, with a particular focus on the Productive Use of Renewable Energy (PURE) in agricultural value chains. The delivered scope comprises three connected areas:

- A public knowledge platform — homepage, project directory, technology catalogue, thematic knowledge hub, ecosystem and partner directory, resource library, events and learning hub, opportunities including tenders and jobs, news and stories, a geographic map, an open-data area with machine-readable downloads, ranked cross-module search, an about page, an accessibility statement, a newsletter and a contact and submission form.
- An administration portal — content management for every content type, an editorial review queue, user and role management, taxonomy and site configuration, a contact inbox, newsletter subscriber management, an audit trail, and an analytics and reporting module with document exports.
- A partner self-service portal — organisation and individual self-registration, an organisation team-management area, content submission into the editorial workflow, and a notifications inbox.

Future phases — provisioned for architecturally but outside the current delivery — include GIS deep-dive tooling, investment matchmaking, and integration with external national digital systems through the platform's public API.

1.3 Definitions & Abbreviations

TERM	DEFINITION
NREP	National Renewable Energy Platform — the platform owner and commissioning body.
PUE / PURE	Productive Use of Energy / Productive Use of Renewable Energy.
SRS	Software Requirements Specification (this document).
ToR	Terms of Reference for the consultancy.
RBAC	Role-Based Access Control.
UAT	User Acceptance Testing.
CMS	Content Management System (the admin portal).
API	Application Programming Interface — here, the platform's REST/JSON interface.
GIS	Geographic Information System.
Ecosystem	The platform's collective term for registered partners, both organisations and individuals; also the public menu under which the partner directory and registration flows sit.
Workflow status	The editorial state of a content record: draft, pending, in review, approved, published, rejected or archived.
Organisation-scoped role	A role that grants powers only over records and accounts belonging to the holder's own partner organisation.

1.4 References

- Terms of Reference: Consultancy Services for the Design, Development, Deployment and Technical Support of the Agri-Energy Desk Digital Platform.
- Requirements Analysis Report (Deliverable 2).
- UI/UX Design Package (Deliverable 4).
- Database Design (Deliverable 6), realised as the migration set in the platform repository.
- Deployment and operations runbook (deploy/README.md in the platform repository).
- Client change requests arising from the design review, consolidated in the project's modification log and reflected throughout Section 3.
- FumbaHub national clean-cooking platform (fumbahub.ug) — sister NREP platform, referenced for institutional and data conventions.

1.5 Revision History

VERSION	DATE	SUMMARY OF CHANGE
1.0	Pre-development	Original specification submitted for approval prior to development.
2.0	August 2026	Revised to the as-built platform. Adds the ecosystem and partner registration flows, the partner portal, events, opportunities, tenders and jobs, news, open data, notifications, analytics and reporting, internationalisation and accessibility. Expands the user classes from three to a nine-way access model, restates the data model against the delivered schema, and adds Sections 8 (security and access control), 10 (implementation status) and 11 (recommendations).

2. Overall Description

2.1 Product Perspective

The Agri-Energy Desk is a standalone, web-based platform built as two cooperating applications. A Laravel REST API owns all data, authentication, authorisation, the editorial workflow and the audit trail, and persists to a PostgreSQL database. A Next.js application renders the public site, the administration portal and the partner portal, and consumes that API. The two communicate only over the versioned /api/v1 JSON interface, so the same interface that serves the platform's own frontend is available to future external consumers without new development.

Public pages are rendered per request against the API, so content approved in the administration portal appears on the public site immediately, with no rebuild or publication step.

2.2 User Classes

The three user classes of Version 1.0 have been replaced by a finer-grained model that separates platform staff from external organisations, and separates an organisation's internal administration from any power over the platform. Access is enforced by role-based access control in the API.

USER CLASS	DESCRIPTION
Public Visitor	Unauthenticated user browsing, searching, filtering and downloading published content, subscribing to the newsletter, and submitting information or enquiries through the public form.

USER CLASS	DESCRIPTION
Super Administrator	Full platform control: system settings, security, the audit trail, backups, user and role management, approvals and all content. Bypasses every authorisation gate.
Platform Administrator	Day-to-day operations: user and organisation management, knowledge resources, events, opportunities, dashboards and reports. May approve and publish content, but may not change core system settings or read the audit trail.
Knowledge Manager	Curates the knowledge repository: reviews and publishes content, organises categories, manages metadata and tags, and upholds quality standards. No user management or system settings.
Content Contributor	Platform staff who create and edit their own content and upload files, submitting it for review. Cannot publish directly.
Partner	An external organisation or individual that has self-registered and been approved. Uses the partner portal, not the administration portal. Authors its own content, which enters review rather than publishing directly.
Partner Administrator	The account that registers an organisation. Administers that one organisation: creates and manages other accounts belonging to the same organisation and assigns them organisation-scoped roles. Holds no platform-wide powers.
Partner Knowledge Manager	An organisation's own knowledge lead. Authors and organises that organisation's content; submits for review, never publishes.
Partner Content Contributor	An organisation's own content author. Creates and edits that organisation's records and uploads files; submits for review.

2.3 Operating Environment

Client side: any current web browser on desktop, tablet or mobile. Server side: a Linux host running PHP 8.2 or later with PHP-FPM for the API, PostgreSQL 14 or later for data, and Node.js 20 or later for the frontend server, with nginx terminating TLS and acting as the single public entry point. The frontend server binds to loopback only and is reached exclusively through nginx, so the platform presents one origin and requires no cross-origin configuration. Uploaded media is served from the same origin.

The current UAT environment runs this stack on a single cloud host; the production topology shall transition directly to the decoupled managed database, cloud object storage, and isolated container architecture specified in Section 7.6

2.4 Design & Implementation Constraints

- Visual identity retains NREP's agri-green brand identity while following an institutional, editorial layout structure consistent with NREP's broader digital presence (FumbaHub).
- The platform is delivered as a modular architecture such that individual modules (for example the GIS map or a future matchmaking module) can be extended post-launch without full redevelopment.
- All data access is through the versioned REST API; the frontend holds no direct database connection, which keeps authorisation decisions in one place.
- Organisation-scoped roles must never be able to grant platform-wide roles. An external organisation administering its own team must not be able to mint an account that can publish to the national platform.
- The public site must remain usable on low-bandwidth mobile connections, which are the dominant access mode for the platform's district-level audience.
- All source code, documentation and design assets are the sole property of NREP on completion, per the ToR's Intellectual Property provisions.

2.5 Assumptions & Dependencies

- NREP supplies or validates the initial content for projects, technologies, partners, resources, events and opportunities ahead of go-live. The content currently loaded in UAT is illustrative demonstration data and is not to be presented as a record of real programmes, budgets or beneficiaries.
- Geocoding of project locations depends on an external place-name lookup service; map base tiles are served by an external tile provider. Both are read-only dependencies and neither holds platform data.
- Outbound email — verification, notifications, newsletter confirmations — depends on a mail relay provisioned by NREP with a sending domain authenticated for the platform.
- NREP nominates the super administrator and the platform administrators, and is responsible for approving partner registrations.

3. System Features (Functional Requirements)

3.1 Public Homepage

ID	REQUIREMENT
FR-1.1	The system shall display platform-wide statistics (project, technology, theme, partner and resource counts, and districts reached).

ID	REQUIREMENT
FR-1.2	The system shall display a curated set of featured projects, technologies, themes, partners and resources, configurable by an administrator through site configuration.
FR-1.3	The system shall provide a prominent search entry point with type-ahead suggestions, available from the homepage and from every page header.
FR-1.4	The system shall display a collaborating-partners strip presenting the platform's institutional collaborators.
FR-1.5	The system shall provide a prominent Create account call to action leading to the registration flow, alongside a route into the partner portal for existing account holders.
FR-1.6	The system shall provide a newsletter sign-up on the homepage.

3.2 Project Directory

ID	REQUIREMENT
FR-2.1	The system shall list all published projects with filters for status, region, district, technology, theme, value chain and implementing partner, and with free-text search.
FR-2.2	The system shall provide a detail page per project showing summary, description, location, coordinates, timeline, status, budget and currency, beneficiary count and beneficiary tags, funding partner, implementing partners, value chain, outcomes, related technologies and themes, an image gallery and downloadable attachments.
FR-2.3	The system shall support projects that span multiple regions and districts, in addition to a primary region and district, with an optional free-text location note.
FR-2.4	Content managers shall be able to create, edit, publish, unpublish and delete project records; contributors and partners shall be able to create and edit their own unpublished records and submit them for review.
FR-2.5	When creating a project, an authorised author shall be able to nominate registered partners — by partner type, such as funding or implementing — to be notified when the record is published.
FR-2.6	Region and district shall be selected from controlled lists rather than typed freely, in both the administration portal and the partner portal, with the district list determined by the selected region.

3.3 Technology Catalogue

ID	REQUIREMENT
FR-3.1	The system shall list all published technologies, filterable by category and theme, with free-text search.
FR-3.2	The system shall provide a detail page per technology covering applications, benefits, limitations, operational notes, maintenance notes, related projects, themes and resources.
FR-3.3	Operational notes shall accommodate deployment and cost detail — including the amount expended and whether the technology used was newly acquired, existing or procured — as structured, repeatable entries rather than a single free-text field.
FR-3.4	Authorised users shall be able to manage technology records including all structured guidance fields.
FR-3.5	Where another record refers to a technology, the reference shall be selected from the registered technology catalogue rather than typed freely.

3.4 Thematic Knowledge Hub

ID	REQUIREMENT
FR-4.1	The system shall present cross-cutting themes, each linking to related projects, technologies and resources.
FR-4.2	Each theme page shall present a why-it-matters summary in addition to its description.
FR-4.3	Themes shall be manageable through the administration portal and selectable as a filter across the project, technology and resource directories.

3.5 Ecosystem: Partner Directory & Registration

The partner area is presented publicly as the Eco System menu, grouping the directory of registered partners with the two routes into it.

ID	REQUIREMENT
FR-5.1	The system shall list all published partner organisations, filterable by organisation type and region, with free-text search.
FR-5.2	The system shall provide a partner profile page with contact details, regional and district coverage, areas of focus, and related projects, technologies and themes.

ID	REQUIREMENT
FR-5.3	The system shall provide self-registration through a guided multi-step wizard, offering two routes: registration as an individual and registration as an organisation.
FR-5.4	Individual registration shall capture full name, email, username, telephone and password.
FR-5.5	Organisation registration shall capture, as required fields: organisation name, registration number, tax identification number, organisation email, physical address, sector of operation, mission, brief description and areas of expertise.
FR-5.6	Areas of expertise shall be a multiple-selection field limited to three entries, with that limit stated to the registrant.
FR-5.7	An organisation shall declare a coverage scope of National or International. Selecting National shall present region and district selection, where choosing a region loads that region's districts; selecting International shall present country selection instead. The two are mutually exclusive and only the selected one shall be shown.
FR-5.8	Organisation registration shall capture a contact person — full name, position, email and telephone — with all fields required and the email pre-filled from the organisation email.
FR-5.9	Account creation shall capture a username, described to organisation registrants as the organisation username, alongside the login email and password.
FR-5.10	An organisation shall declare a partner type, distinguishing at least funding partner from implementing partner.
FR-5.11	Registration shall additionally capture optional profile detail: acronym, year established, postal address, value chains, employee, branch, beneficiary and annual budget bands.
FR-5.12	A registration shall create an account with pending status and, for organisations, a draft partner profile. Neither the account nor the profile shall be active or publicly visible until an administrator approves it.
FR-5.13	The system shall verify a registrant's email address through a signed verification link.
FR-5.14	The partner sign-in page shall be presented simply as Sign in.

3.6 Resource Library

ID	REQUIREMENT
FR-6.1	The system shall list all published resources, filterable by resource type and theme, with free-text search.
FR-6.2	The system shall support a file attachment and/or an external link for each resource, together with a publication date and related themes and technologies.
FR-6.3	When an authorised user attaches a file to a resource, the system shall provide an access setting distinguishing view-only material from material that may also be downloaded.
FR-6.4	A resource marked view-only shall remain readable through the platform but shall not present a download action; the setting shall be available to both administration and partner authors.

3.7 Events & Learning Hub

ID	REQUIREMENT
FR-7.1	The system shall list published events — conferences, workshops, webinars and training — with a detail page for each.
FR-7.2	An event shall record its type, start and end date and time, whether it is online, its location, a registration link and a featured image.
FR-7.3	Events shall be manageable through the administration portal and submittable by partners through the partner portal.

3.8 Opportunities, Tenders & Jobs

ID	REQUIREMENT
FR-8.1	The system shall list published opportunities with a detail page for each, filterable by type, sector, value chain and region.
FR-8.2	An opportunity shall be classified by type: funding, investment, tender, procurement, or job and vacancy.
FR-8.3	An opportunity shall record its funder, amount and currency, sector, value chain, region, application deadline, eligibility criteria and an external application link.
FR-8.4	The system shall provide a dedicated tenders and procurement browser over opportunities of the procurement-related types.
FR-8.5	Opportunities shall be manageable through the administration portal and submittable by partners through the partner portal.

3.9 News & Stories

ID	REQUIREMENT
FR-9.1	The system shall publish news items, field stories, press releases and announcements, each with an excerpt, body, featured image, gallery and publication date.
FR-9.2	The system shall provide a news index and a detail page per item.

3.10 Geographic Map & Open Data

ID	REQUIREMENT
FR-10.1	The system shall display an interactive map of Uganda showing project locations, with the whole country within the initial viewport and no clipping of the national boundary.
FR-10.2	The map shall occupy a viewport large enough to read district-level detail without immediate zooming, and the map area shall not be reduced by a redundant list of the projects already visible on it.
FR-10.3	The system shall allow users to filter the map by region, by district and by project status, and to clear all filters in one action.
FR-10.4	Selecting a project on the map shall present its summary details with a link to its detail page.
FR-10.5	The map page shall present its statistics as charts rather than plain figures, including a project status breakdown and a comparative view of totals by region, and the charts shall respond to the active filters.
FR-10.6	The system shall provide a public open-data area presenting regional aggregates — project counts, budget totals and beneficiary figures — as charts.
FR-10.7	The system shall publish machine-readable downloads of the project dataset in CSV and GeoJSON, and shall document the public read endpoints from which they are drawn.
FR-10.8	The administration and partner portals shall validate that project geocoordinates fall strictly within the bounding coordinates of Uganda (**Latitude: -1.5° to 4.3° N; Longitude: 29.5° to 35.0° E**), throwing a validation error for any inputs outside this threshold.
FR-10.9	The map shall implement spatial point-clustering at regional and district zoom levels to optimize client-side rendering performance and memory consumption on low-bandwidth mobile devices.

3.11 Search

ID	REQUIREMENT
FR-11.1	The system shall provide a cross-module search returning ranked results across projects, technologies, themes, partners, resources, events, opportunities, policies and news.
FR-11.2	Search shall return per-type result counts so users can narrow results to a content type.
FR-11.3	Search shall support filtering by thematic area, technology, organisation, geographic location and publication type.
FR-11.4	The system shall offer type-ahead suggestions from the header search field.
FR-11.5	Search shall only ever return published records.

3.12 Contact & Public Submission

ID	REQUIREMENT
FR-12.1	The system shall provide a public form for submitting new project, technology, partner or resource information, or a general enquiry.
FR-12.2	Submissions shall be stored with a status (new, read, responded, archived) and be visible to authorised staff in the administration portal, with a record of whether an acknowledgement was sent.
FR-12.3	Staff shall be able to record internal notes against a submission and update its status.
FR-12.4	The system shall provide a newsletter subscription with a confirmation step, and a one-click unsubscribe link in every issue.

3.13 Partner Portal

ID	REQUIREMENT
FR-13.1	The system shall provide an authenticated portal, separate from the administration portal, in which an approved partner manages its own contributions.
FR-13.2	A partner shall be able to submit projects, technologies, resources, events and opportunities, and to see the workflow status of everything it has submitted.

ID	REQUIREMENT
FR-13.3	Partner submission forms shall capture the same fields as the equivalent administration forms, so that a partner-authored record is not missing information an administrator would have recorded.
FR-13.4	A partner administrator shall be able to create and manage additional accounts within its own organisation and assign them organisation-scoped roles, and shall be able to suspend and restore those accounts.
FR-13.5	A partner administrator shall not be able to grant any platform-wide role, and shall not be able to see or act on accounts belonging to another organisation.
FR-13.6	The portal shall provide a notifications inbox in which a partner sees platform notifications addressed to it, with read and read-all actions.

3.14 Editorial Workflow & Notifications

ID	REQUIREMENT
FR-14.1	Every content record shall carry a workflow status: draft, pending, in review, approved, published, rejected or archived.
FR-14.2	Content authored by a contributor or a partner shall enter the workflow as pending and shall never publish directly.
FR-14.3	The system shall provide a review queue listing everything awaiting review across all content types, with the record's type, author and submission date.
FR-14.4	An authorised reviewer shall be able to approve or reject a submission, recording review notes and the time of review; a rejection shall return the record to its author with those notes.
FR-14.5	On publication, the system shall notify the partners nominated on the record, both in their portal notifications inbox and by email.
FR-14.6	Every content and user-management action shall be written to an audit trail recording the actor, the action, the affected record and the time.

3.15 Administration Portal

ID	REQUIREMENT
FR-15.1	The system shall require authentication for all administration portal access, with role-based access control across the roles defined in Section 2.2 and enforced in the API.
FR-15.2	The system shall provide content management screens for every entity in Section 6, including create, edit, publish, unpublish and delete actions.

ID	REQUIREMENT
FR-15.3	Every administration list shall provide filters for rapid action, including free-text search, filtering by status and by other record attributes such as type, category, role or organisation, and filtering by a date range, with a single action to clear all filters and a visible count of matching records.
FR-15.4	The system shall provide a submissions queue with review, status update and response tracking.
FR-15.5	The system shall provide user management: creating accounts directly from the administration portal, approving pending registrations, suspending and restoring accounts, and assigning or removing roles.
FR-15.6	Only a super administrator shall be able to grant the super administrator role or alter the roles of an account that holds it; the interface shall reflect the same restriction the API enforces.
FR-15.7	The system shall provide taxonomy management (value chains, project statuses, resource types, partner types, technology categories, beneficiary tags), with regions and districts drawn from the national geographic reference.
FR-15.8	The system shall provide site configuration management (hero content, headline statistics, featured item selection, contact details, about summary).
FR-15.9	The system shall provide newsletter subscriber management, including export of the subscriber list.
FR-15.10	The system shall provide an audit-trail view, restricted to the super administrator.
FR-15.11	The system shall support file and image upload for content records, restricted to approved authors.
FR-15.12	All data tables, lists, and filtering operations within the administration and partner portals shall execute via server-side pagination and processing once an entity's record volume surpasses 1,000 entries.

3.16 Analytics & Reporting

ID	REQUIREMENT
FR-16.1	The system shall provide an analytics overview summarising content by type, workflow state and publication state, together with user, partner and submission totals and recent activity.
FR-16.2	The system shall provide a reporting module producing period-scoped reports, comprising at least: platform overview, content inventory, content

ID	REQUIREMENT
	growth and trends, content utilisation, user engagement, project portfolio, stakeholder directory, and opportunities and events.
FR-16.3	Each report shall be self-describing — stating its period, summary figures and tabular sections — so that it can be rendered or exported without report-specific handling.
FR-16.4	The system shall allow reports and analytics extracts to be exported as PDF, Excel and CSV documents.
FR-16.5	Analytics and reporting shall be restricted to content managers and above.

3.17 Internationalisation & Accessibility

ID	REQUIREMENT
FR-17.1	The public interface shall be available in English, French and Kiswahili, with English as the default.
FR-17.2	A visitor shall be able to switch language from any page, and the choice shall persist across visits without altering page addresses.
FR-17.3	The system shall provide a skip-to-content link, keyboard-operable navigation, and accessible names for interactive controls.
FR-17.4	The system shall publish an accessibility statement describing the standard targeted and how to report a barrier.
FR-17.5	The localized application routing structure shall feature an automated translation fallback hierarchy. If a translation key is missing or unverified within the French or Kiswahili dictionaries, the system shall cleanly render the default English string without breaking layouts.

4. External Interface Requirements

4.1 User Interfaces

Responsive web interface covering desktop, tablet and mobile breakpoints, delivered per the approved UI/UX Design Package (Deliverable 4). The public site, the administration portal and the partner portal share one design system, one component library and one navigation grammar, so that a user moving between them is not learning a second interface.

4.2 API Interfaces

A versioned REST API under /api/v1 exposes unauthenticated read access to published projects, technologies, themes, partners, resources, events, opportunities, policies and news, together with

taxonomy, site configuration, search and open-data aggregates. Authenticated write access covers content creation and editing, the editorial workflow transitions, user and team management, submissions, newsletter administration, notifications, analytics, reporting and the audit trail. Authentication uses bearer tokens issued on sign-in. The interface is documented as a browsable OpenAPI specification generated from the implementation, which is the reference for any future external integration.

4.2.1 Uniform Error Architecture

To ensure seamless troubleshooting and clean failure mapping for current and future external consumers, the API shall return a predictable error envelope for all non-2xx HTTP responses. Every response handling an exception, validation failure, or authentication denial shall structure its payload using the following rigid contract:

json

```
{
  "success": false,
  "error": {
    "code": "VALIDATION_FAILED",
    "message": "The provided coordinates point outside the national boundary of Uganda.",
    "details": {
      "coordinates": ["The latitude must be between -1.5 and 4.3."]
    },
    "timestamp": "2026-09-21T17:23:00Z"
  }
}
```

4.3 Email & Notification Interfaces

The platform sends transactional email through an SMTP relay: address verification at registration, newsletter confirmation and unsubscribe, and publication notifications to nominated partners. In-platform notifications are delivered to the recipient's portal inbox independently of email, so a mail delivery failure does not lose the notification.

4.4 Third-Party Service Interfaces

Two external read-only services are used at the presentation layer: a place-name geocoding service, which resolves typed locations to coordinates when a record is authored, and a map tile provider, which supplies the base cartography for the interactive map. Neither holds platform data, and the platform stores the resulting coordinates itself so that map rendering does not depend on the geocoder at read time.

5. Non-Functional Requirements

CATEGORY	REQUIREMENT
Performance	The platform shall maintain a strict page-weight budget maximizing at **250KB (zipped)** for critical assets on public views. The API shall respond to read queries within **200ms** under a simulated load of 500 concurrent requests. The Next.js frontend shall target a minimum Lighthouse performance score of **90/100** on simulated mid-tier mobile devices using 3G throttling. Search queries shall utilize full-text database ranking vectors to guarantee completion times under **100ms** regardless of search string length..
Availability	The platform shall achieve high availability with minimal downtime, supported by automated backups of the database and uploaded media, and documented, rehearsed recovery procedures.
Security	The platform shall require Multi-Factor Authentication (MFA) for all Super Administrator and Platform Administrator accounts prior to admin portal access. The REST API shall enforce a rate limit of 60 requests per minute per IP address on authentication, registration, upload, and public submission routes, returning an HTTP 429 Too Many Requests response when breached. Bearer tokens shall expire after 15 minutes and require a secure, HTTP-only refresh token to rotate
Data protection	In strict accordance with Uganda's Data Protection and Privacy Act, all PII collected via registration, forms, and newsletters shall explicitly log the user’s timestamped consent. Public submissions and contact entries shall be systematically anonymized or expunged after 24 months of systematic inactivity.
Usability	The interface shall be intuitive and consistent for all user classes, following the approved design system, with controlled lists in place of free text wherever a controlled list exists.
Accessibility	The interface shall be operable by keyboard, shall provide accessible names for controls, and shall target WCAG 2.1 Level AA.

CATEGORY	REQUIREMENT
Internationalisation	All public interface text shall be externalised into message catalogues so that further languages can be added without code changes.
Compatibility	The platform shall function correctly on current versions of major desktop and mobile browsers, and shall remain usable on low-bandwidth mobile connections.
Scalability	The architecture shall support growth in content volume and future modules without major redevelopment. Administration screens shall move to server-side pagination and filtering as record volumes grow.
Maintainability	Code shall be documented and modular. Automated tests shall cover authentication, access control, the editorial workflow, notifications, search and reporting, and shall run before each release.
Auditability	It shall be possible to establish, for any published record, who created it, who approved it and when.

6. Data Model (Core Entities)

The following entities and key attributes describe the relational schema in use. Publishable entities additionally share a common set of editorial columns — featured flag, published flag, workflow status, author, review notes and review timestamp — and are related to one another through many-to-many association tables, so that a project, a technology, a theme, a partner and a resource can each reference the others without duplication.

ENTITY	KEY ATTRIBUTES
Project	slug, title, summary, description, region, district, multi-region and multi-district coverage, location note, coordinates, funding partner, budget and currency, beneficiary count, beneficiary tags, value chain, status, start and end date, outcomes, featured image, gallery, attachments, external link, partners to notify on publication; related technologies, themes and partners (with implementing or funding role)
Technology	slug, name, category, summary, description, applications, benefits, limitations, operational notes, maintenance notes, featured image; related projects, themes and resources
Theme	slug, name, icon, summary, description, why-it-matters, featured image; related technologies, projects and resources

ENTITY	KEY ATTRIBUTES
Partner	slug, name, acronym, type, category, summary, description, website, email, phone, logo, registration number, tax identification number, year established, physical and postal address, coverage scope, regions, district coverage, countries, sectors, value chains, mission, expertise, employee, branch, beneficiary and budget bands, contact person name, position, email and phone; related technologies, themes and projects
Resource	slug, title, type, summary, description, file, download-permitted flag, external link, featured image, publication date; related themes and technologies
Event	slug, title, type, summary, description, start and end date-time, online flag, location, registration link, featured image
Opportunity	slug, title, opportunity type, funder, summary, description, amount, currency, sector, value chain, region, deadline, eligibility, external link
Policy	slug, title, category, summary, description, document link, publication date
Post	slug, title, category, excerpt, body, featured image, gallery, publication date
Submission	submission type, name, organisation, email, subject, message, status, acknowledgement status, admin notes, submitted and updated timestamps
Subscriber	email, confirmation state, confirmation and unsubscribe tokens, subscription timestamps
Notification	type, recipient, payload, read timestamp
Taxonomy	grouped controlled lists: value chains, project statuses, resource types, partner types, technology categories, beneficiary tags; with regions, districts and sub-counties from the national geographic reference
Site configuration	hero content, headline statistics, featured item selections per content type, contact email and phone, about summary
User	name, username, email, hashed credentials, status, account type (individual or organisation), organisation, phone, linked partner, approval timestamp, roles
Audit entry	actor, action, affected record, changed attributes, timestamp

6.1 Relational Schema & Integrity Constraints

- **Soft Deletions:** To protect audit trails and historical records, the database shall implement logical soft-deletions via a deleted_at timestamp for all core content types (Projects, Technologies, Resources, Partners). Hard DELETE actions are strictly forbidden at the application level.
- **Cascade Restraints:** The relational schema shall enforce ON DELETE RESTRICT constraints on all foreign key pairings governing controlled taxonomies. The system shall block an administrator from deleting a district, region, or value chain if it is currently tied to a published or draft content record.
- **File Layer Restrictions:** View-only files specified under requirement FR-6.3 shall be decoupled from direct public URL access. The server shall serve these files through a protected API endpoint that generates short-lived signature tokens, blocking downstream file pulling outside the platform's visual component.

7. System Architecture Overview

The platform is a two-tier web application. The backend is a Laravel REST API over PostgreSQL, holding all data, authentication, role-based authorisation, the editorial workflow, full-text search, media handling, analytics, reporting and the audit trail. The frontend is a Next.js application rendering the public site, the administration portal and the partner portal, and consuming only the versioned API.

In deployment, nginx terminates TLS and is the single public entry point. It routes API, media and health paths to PHP-FPM and everything else to the Node.js frontend server, which binds to loopback only. Because both tiers are served from one origin there is no cross-origin configuration to maintain, and the frontend's own server-side rendering calls never leave the host. Public pages render per request, so publication is immediate.

Authorisation is centralised: a small set of named gates and a shared content policy express every access decision, and the role groups they consult are defined in a single place. Adding a role or changing what a role may do is therefore a change in one file, not a search through controllers.

The architecture reserves clear extension points for a deeper GIS module, investment matchmaking and external system integration, consistent with the ToR's scalability requirements. Because the public read API already exposes the full published dataset, an external consumer can be onboarded without new backend development.

7.1: Future Architecture & Extensibility Hooks

7.1.1 Modular Isolation Policy

To guarantee that Phase 2 expansions do not destabilize or break the core platform built under Version 2.0, the system enforces a strict isolation boundary between the core platform application blocks and upcoming components.

All future services **GIS Deep-Dive Tools, Investment Matchmaking, the PURE Marketplace, and AI Insights** shall interface with the existing codebase exclusively via decoupled Laravel Service Providers, localized Event Listeners, and the outbound /api/v1 gateway. The base database tables for Core Projects, Technologies, and Users shall remain immutable to direct schema modification by external scripts.

7.1.2 Geographic Information System (GIS) Deep-Dive Integration

- **Data Hook:** Core projects currently store explicit coordinates under requirement FR-2.2. The platform exposes these via a standardized GeoJSON stream loop /api/v1/open-data/geojson.
- **Extension Boundary:** The upcoming GIS deep-dive module shall run as an independent micro-frontend within the Next.js topology. It shall ingest the existing open-data GeoJSON endpoint rather than querying the database directly.
- **Write-Back Vector:** Advanced spatial layers, catchment analyses, and agricultural overlay models generated by the GIS engine shall be mapped back to core projects using a singular, foreign indexing schema tied entirely to the project slug attribute.

7.1.3 Investment Matchmaking Component

- **Data Hook:** Registered partners are already split into standardized buckets, including "Funding Partner" and "Implementing Partner" under FR-5.10.
- **Extension Boundary:** Matchmaking workflows shall leverage an asynchronous Event-Driven pipeline. When an implementing partner updates a project profile seeking financing, the core API shall fire a native Laravel system event (ProjectSeekingCapital).
- **Isolation Guard:** The Matchmaking database engine shall run on localized tables separate from user account records. It reads target parameters through an isolated mapping service provider, protecting core identity data from structural exposure.

7.1.4 Productive Use of Renewable Energy (PURE) Marketplace Component

- **Data Hook:** The system already maps and references a strict Taxonomy matrix comprising verified technology categories, value chains, and regional boundaries.

- **Functional Scope (Phase 2 Add-On):** The Marketplace module shall introduce verified peer-to-peer listings allowing vendors to advertise PURE equipment (e.g., solar water pumps, solar dryers, cold storage systems) directly to agricultural cooperatives.
- **Architecture Mapping:** Marketplace items shall tie directly into the existing Technology catalogue items (FR-3.5) and Partner entities (FR-5.2) using Laravel **Polymorphic Relations**. This ensures vendors cannot alter the global technology baseline records, but can link inventory instances cleanly to their approved profiles.

7.1.5 Artificial Intelligence (AI) & Intelligence Automation Layer

- **Vector Engine Hook:** The database schema shall accommodate a fallback vector data type (via pgvector on the existing PostgreSQL instance). This maps text inputs from the Resource Library, Technology Catalogue, and Project records into multi-dimensional embeddings without disrupting standard tabular reads.
- **AI Smart Search & Matching:** An isolated AI orchestration service shall hook into the /api/v1 layer to:
    - Expose semantic natural-language search over the core indices, supplementing the literal full-text database filters.
    - Automatically scan incoming partner project proposals and match them against known eligibility matrices inside published open opportunities (FR-8.3).
    - Auto-generate translated content summaries for native-speaker verification teams processing French and Kiswahili localizations.

7.2: Production Deployment Topology & Cloud Infrastructure

7.2.1 Cloud-Native Architecture Overview

While the User Acceptance Testing (UAT) environment is deployed on a single virtual private server to minimize costs during review, the production environment shall utilize a decoupled, highly available cloud infrastructure layout. The platform is architected to isolate compute, data persistence, and asset storage layers. This layout allows individual scaling of the Next.js frontend, the Laravel API, and future resource-heavy modules (AI vectors and GIS computation) without creating resource competition.

7.2.2 Compute & Scaling Layer

- **Decoupled Clusters:** The frontend Next.js application server and the backend Laravel API engine shall run in isolated execution environments (e.g., containerized pods or decoupled virtual instances) behind an **Application Load Balancer (ALB)**.

- **Stateless Execution:** Both compute layers shall remain completely stateless. User sessions and bearer tokens are checked statelessly via JWT or database-backed cache records, allowing instances to scale horizontally automatically during sudden traffic spikes (e.g., during national tender releases or agricultural funding announcements).

7.2.3 Data & Storage Persistence Isolation

- **Managed Database Layer:** The platform shall utilize a managed database instance running **PostgreSQL 14+ with the pgvector extension** active. Production storage must deploy automated daily snapshotting, multi-availability zone replication for disaster recovery, and point-in-time recovery (PITR) logs.
- **Decoupled Object Storage:** In adherence to recommendation 11.3, all user-uploaded files, technology maintenance files, project images, and downloadable CSV/GeoJSON datasets shall be immediately offloaded to a secure, distributed **Cloud Object Storage Service (such as AWS S3 or equivalent)**. Compute nodes shall not store local content assets.
- **Media Delivery CDN:** Static frontend assets and public file downloads shall be distributed globally via an **Edge Content Delivery Network (CDN)**, ensuring high-speed access for low-bandwidth mobile devices across regional districts in Uganda while reducing origin load.

7.2.4 Phase 2 Isolation Gates (GIS, Marketplace, & AI Engines)

To prevent advanced data features from interrupting daily operations, production infrastructure boundaries shall be strictly enforced at the resource layer:

- **AI Compute Sandbox:** High-latency semantic operations, vector search embedding updates, and automated translation processes shall execute on isolated asynchronous worker containers (queues managed by Redis or Amazon SQS). They will never execute on the HTTP web server thread handling public site visits.
- **GIS Engine Data Caching:** Heavy geospatial computation and spatial point calculations shall utilize an in-memory cache layer (**Redis**). This prevents intensive geographic lookups from exhausting the primary relational database's connection limits.
- **Marketplace Sandbox:** Transaction logging, product inventory lookups, and vendor-partner interaction histories shall live in isolated database tables. These will use a dedicated read-replica instance to prevent read/write locks from impacting the core national directory views.

8. Security & Access Control

Access is enforced in the API on every request; the interface reflects those rules but is never the thing that enforces them. The table below summarises what each role may do. Organisation-scoped roles apply only within the holder's own organisation.

ROLE	CONTENT	ADMINISTRATION
Super administrator	All content; publish; review queue	Everything, including system settings and the audit trail
Platform administrator	All content; publish; review queue	Users, roles, organisations, taxonomy, site configuration, analytics and reports; not system settings or the audit trail
Knowledge manager	All content; publish; review queue	Taxonomy and metadata; analytics and reports; no user management
Content contributor	Own content only, submitted for review	None
Partner	Own content only, submitted for review	None; partner portal only
Partner administrator	Own organisation's content, submitted for review	Own organisation's accounts and organisation-scoped roles only
Partner knowledge manager	Own organisation's content, submitted for review	None
Partner content contributor	Own organisation's content, submitted for review	None
Public visitor	Read published content only	None

- Authentication issues a bearer token on sign-in; the token is presented on every subsequent API call and can be revoked by signing out.
- Registration, sign-in, upload and public-form endpoints are rate limited to blunt brute-force and automated abuse.
- A newly registered account is inactive until an administrator approves it, and a suspended account loses access immediately.

- System settings and the audit trail are reachable by the super administrator alone; the corresponding gates deny every other role by construction rather than by an omitted check.
- Uploads are restricted to approved authors and are served from a path that does not execute server-side code.
- **Malicious Execution Countermeasures:** All media assets submitted through user uploads (FR-15.11) shall be routed through a dedicated validation filter that strips alternative file extensions and evaluates strict MIME types rather than trusting user-supplied headers. Files shall be stored on a path where the web server configuration explicitly disables execution flags (e.g., stripping execution headers for .php, .js, or .sh variants), forcing browsers to view them exclusively as static binary streams or media attachments.

9. Traceability Summary

TOR OBJECTIVE	ADDRESSED IN
Secure, scalable, responsive platform	Sections 5, 7, 8
Integrated knowledge management system	Sections 3.2–3.9
Digital collaboration tools across stakeholders	Sections 3.5, 3.12, 3.13, 3.14
Administrative tools for content, user and resource management	Sections 3.14, 3.15
Search, analytics and reporting	Sections 3.11, 3.16
Geographic visualisation and open data	Section 3.10
Inclusive, multilingual public access	Section 3.17
Future GIS, API and integration readiness	Sections 2.4, 4.2, 7

10. Implementation Status Summary

The status of each module against this specification, as at this revision.

MODULE	STATUS	NOTES
Public knowledge platform	Implemented	All directories, detail pages, map, open data, search, news, events and opportunities live in UAT.
Ecosystem registration	Implemented	Individual and organisation wizards, national and international coverage, approval gate.
Administration portal	Implemented	Content management, filters, user creation and approval, taxonomy, site configuration, submissions inbox.
Partner portal	Implemented	Submission of five content types, team management, notifications inbox.
Editorial workflow	Implemented	Seven-state workflow, review queue, approve and reject with notes, audit trail.
Notifications	Implemented for projects	Publication notification is currently raised from project records only; extension to the other content types is recommended in Section 11.
Analytics & reporting	Implemented	Eight reports with PDF, Excel and CSV export.
Internationalisation	Implemented	English, French and Kiswahili catalogues; French and Kiswahili await native-speaker review.
Policy library	Backend complete; no public page	Policies are managed in the administration portal and exposed by the API, but are not yet surfaced as a public section.
Content	Demonstration data	Present content is illustrative and must be replaced with NREP-validated records before go-live.
GIS deep dive, matchmaking, external integrations	Not in this phase	Architecturally provisioned; see Section 7.

MODULE	STATUS	NOTES
PURE Marketplace, AI, GIS Deep-Dive & Matchmaking	Not in this phase (Architecturally Hooked)	Vector-ready DB configurations and GeoJSON read loops are baseline-active in the v1 core API build. Future services drop cleanly into these listening gates.

11. Recommendations

The following are put to NREP for consideration. They are not defects in the delivered scope; they are the changes that would most improve the platform's usefulness, resilience and readiness for public launch. They are grouped by the decision each one needs.

11.1 Before Go-Live

- Replace the demonstration content. Every project, budget, beneficiary figure and partner record currently in the system is illustrative. Publishing the platform with this data would misrepresent the sector, so a content load of NREP-validated records should be treated as a launch gate rather than a follow-on task.
- Enforce view-only resources at the file layer. The download restriction is currently applied in the interface: a view-only file is not offered for download, but its address remains directly reachable by anyone who has it. Restricted files should be served through an access-checked endpoint issuing short-lived links, so that the restriction holds outside the page that presents it.
- Complete the translation review. The French and Kiswahili catalogues are in place and switchable, but should be reviewed by native speakers with sector vocabulary before the language switcher is promoted publicly.
- Publish a privacy notice and define retention periods for registration data, public submissions and newsletter subscriptions, in line with Uganda's Data Protection and Privacy Act.
- Rehearse a restore. Backups are configured; what has not yet been demonstrated is a timed restore of database and uploaded media into a clean host. Recovery time should be measured once and recorded, not assumed.
- Add search-engine essentials — a generated sitemap and per-page metadata — so that published projects, technologies and resources are discoverable from outside the platform. For a national knowledge platform, most first visits will begin at a search engine rather than the homepage.

11.2 Functional Enhancements

- Extend publication notifications beyond projects. The nominate-and-notify mechanism is valuable and currently reaches only project records; applying it to resources, events and opportunities would make the collaboration loop consistent across the platform.
- Surface the policy library publicly. Policies are already managed and served by the API; a public section would complete the knowledge offering at low cost.
- Notify authors on rejection as well as publication. A contributor whose submission is returned currently discovers it by revisiting the portal; an email carrying the reviewer's notes would close the loop.
- Add saved searches and a subscription to new content matching a filter for example new opportunities in a district, or new resources under a theme. This converts the platform from a place people remember to visit into one that reaches them.
- Record resource and document engagement views and downloads per record so that the content utilisation report reflects actual demand rather than inventory.
- Provide bulk import for projects and partners. The first content load, and each subsequent partner onboarding round, would otherwise be entered one record at a time.
- Add a public sector-statistics or dashboard view built on the same aggregates as the open-data area, aimed at policy audiences who want the national picture rather than an individual project.

11.3 Technical & Operational

- Move administration list filtering to the server. Filters currently operate over the records already loaded into the browser, which is fast and simple today but will not hold as content grows. Server-side pagination, filtering and sorting should be introduced before record counts reach the low thousands.
- Materialise the search index. Full-text ranking is computed per query; storing a search vector per record with a matching index would keep search responsive as the corpus grows, without changing behaviour.
- Move uploaded media to object storage in production. Media currently sits on the application host, which couples content to a single machine and complicates both backup and horizontal scaling.
- Scan uploads for malware. The platform accepts files from external organisations and serves them to the public, which is a route worth closing before wide adoption.

- Require multi-factor authentication for super administrator and platform administrator accounts. These accounts can publish to a national platform and manage every other account; a password alone is a thin defence for that level of reach.
- Add end-to-end tests for the critical journeys, registration and approval, submission through review to publication, and search to complement the existing backend test suite and frontend type checking.
- Introduce application monitoring and error reporting with alerting, so that a failure is known before it is reported by a user.
- Add privacy-respecting usage analytics. The reporting module measures content; it cannot yet answer which pages the public actually uses, which is the question NREP will be asked about the platform's reach.

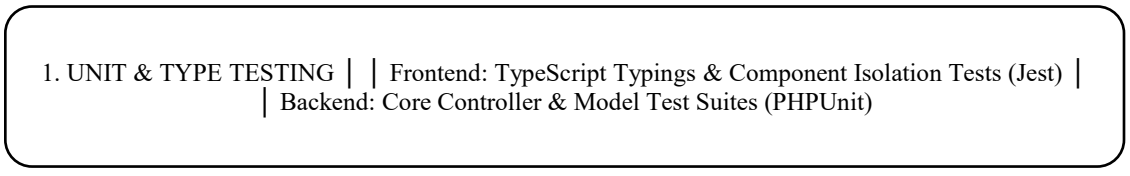
11.4 Governance & Adoption

- Publish an editorial policy stating what qualifies for publication, who reviews what, and the expected turnaround for a partner submission. The workflow is built; its service standard is a decision for NREP.
- Deliver role-based training and a short administrator handbook covering the review queue, user approval and the reporting module, and hand these over with the source code.
- Nominate at least two super administrators. A single holder of the highest-privilege account is an operational risk independent of the software.
- Schedule a content review cycle. A knowledge platform's credibility depends on records not going stale; a periodic review of project status and opportunity deadlines should be assigned rather than left to arise.
- Plan the next phase against the extension points already in place GIS deep dive, investment matchmaking, and integration with other national systems through the existing public API so that Phase 2 begins from the current architecture rather than alongside it.

13. Quality Assurance & Testing Framework

13.1 Testing Strategy Overview

To ensure compliance with the strict performance, security, and extensibility requirements of the Agri-Energy Desk Digital Platform, testing must be executed across both application layers (Next.js and Laravel) independently, followed by end-to-end (E2E) integration checks. This framework isolates testing into four distinct gates to guarantee that Phase 1 core features remain functional when Phase 2 components (AI, GIS, Marketplace) are deployed.



2. ROLE & SECURITY VERIFICATION │ │ API Gateway Authorization Gates & RBAC Route Lockdowns (Postman/Insomnia) │ │ Malware Upload Filters & XSS Script Injection Testing

3. END-TO-END (E2E) INTEGRATION │ │ Simulated Multi-Step Partner Registration, Review Queue & Approvals │ │ Cross-Module Universal Search Vector Assertions (Cypress/Playwright)

4. PERFORMANCE & MOBILE SLA TESTS │ │ 3G Network Emulation & Mobile Lighthouse Threshold Gates (>90/100) │ │ Concurrent REST API Stress Testing under 500-User Loads (JMeter/K6)

13.2 Testing Matrix

Test ID	***Target Module / SLA***	***Testing Method***	***Success Criteria / Assertions***
QA-RBAC.1	Section 2.2 / 8 Role Isolation	Automated API Route Scans (PHPUnit / Postman)	Requests using a Partner or Partner Admin token attempting platform-wide updates must be met with an immediate HTTP 403 Forbidden response.
QA-SEC.2	Section 8 / 11.3 Malware Shield	Binary Injection Testing	Uploading an active script or executable disguised with a .jpg or .pdf extension must fail API validation at the MIME-type step and reject storage.
QA-GEO.3	Section 3.10 / 10.8 Map Validation	Edge-Case Coordinate Injection	Submitting geocoordinates pointing outside Uganda (e.g., Nairobi or London) inside a project payload must trigger a VALIDATION_FAILED API error.

QA-PERF.4	Section 5 / 7.6 API Latency	Asynchronous Load Benchmarking (K6 / JMeter)	Database read endpoints, including /api/v1/projects, must maintain response times below **200ms** under 500 concurrent virtual requests.
QA-MOB.5	Section 2.4 / 5 Low-Bandwidth	Emulated 3G Device Profiling (Lighthouse)	The public Next.js homepage and directory modules must register a Lighthouse performance score ≥ **90/100** under 3G network throttling.
QA-LANG.6	Section 3.17 / 17.5 Localization Fallback	String Dictionary Null-Value Testing	Removing key Kiswahili text blocks from the local catalog file must force the UI to cleanly fall back to English without layout breaking.
QA-EXT.7	Section 7.5 Phase 2 Isolation	System Event Simulation	Firing a mock ProjectSeekingCapital matchmaking hook must execute fully in the background without locking or slowing the active core database tables.

13.3 Test Environment Controls

- **Data Sanitization:** Testing inside the User Acceptance Testing (UAT) environment must use randomized seed content. No real user emails, PII, or actual partner credentials may be used for automated loop checks.
- **Continuous Integration (CI):** All Unit and Type tests (QA-RBAC.1, QA-GEO.3, and QA-LANG.6) must execute automatically on the repository pipeline before code branches can merge into main delivery releases.

Final Step: Document Execution Check

- By compiling all the elements provided throughout this session, your **Agri-Energy Desk Digital Platform SRS** is now upgraded with:
- Formalized enterprise-grade metrics for **Performance** and **Security** (Section 5).
- Advanced constraints for data handling, soft deletions, and secure view-only file layers (Section 6).
- Uniform API exception schemas for seamless external consumer onboarding (Section 4).

- Explicit **Phase 2 Modular Hooks** detailing exact system pathways for **AI**, **GIS**, and the **PURE Marketplace** without code breaking (Section 7.5).
- A cloud-native **Production Infrastructure Model** tracking horizontal auto-scaling and Redis queue background worker sandboxes (Section 7.6).
- A definitive **QA Test Matrix** ensuring complete tracking and verification before final system sign-off (Section 13).

14. Approval

This revision of the SRS is submitted for NREP's formal review and approval as the current definition of the Agri-Energy Desk Digital Platform. Approval confirms the requirements in Sections 3 to 9 as the accepted specification, and records NREP's decisions on the recommendations in Section 11.

APPROVED BY (NREP CONTRACT MANAGER)	SIGNATURE	DATE